Summer 2023
EEE 111L / ETE 111L
Analog Electronics Lab I (Sec-15)
Faculty : Dr. Hafiz Imtiaz (HAI)
Instructor: Md. Rabiul Karim Khan

Lab Report 02:

Study of different diode rectifier circuit

Date of Performance :	**<u>Group Number</u> : 07**
	1.Srabon Das Id: 2232048643
	2.Fariya Jahan Borsha Id: 2233136643
Date of Submission :	3.Halimatus Sadia Id: 2131989642
	4.Anindita Das Mishi Id: 2211364642
	5. Jamea Hena Id: 2222781045

Name of the Experiment: Diode rectifier circuit.

Objective: Study of different diode rectifier circuits.

Theory: A rectifier an electrical device which converts an alternating current into a direct one by allowing a current to flow through it in one direction only. The key component in any rectifier circuit is naturally the diode or diodes used. Rectifier converts an AC signal into a DC signal.
Diode rectifiers can be categorized in two major types. They are –
1. Half-wave rectifier
2. Full-wave rectifier

Half-wave rectifier: It can be built by using single diode. Half-wave rectifier have some major disadvantages.

Full-wave rectifier: Two circuits are used as full-wave rectifier. They are

a) Full-wave rectifier using center tapped transformer - it has some advantages over full-wave rectifier.

b) Full-wave bridge rectifier - it overcomes some disadvantages. Four diodes connected as bridge connection. So, it produces some ripple in the output.

This ripple can be reduce by using filter capacitor across the load.

Equipments and components:

1. p-n junction diode (IN4007) - 4pieces
2. Resistor (10kΩ) - 1 piece
3. Capacitor (0.22 µF, 10 µF) - 1 piece each
4. Signal generator - 1 unit
5. Trainer Board - 1 unit
6. Digital Multimeter - 1 unit
7. Oscilloscope - 1 unit
8. Cords and wires. - as required

Circuit Diagram:

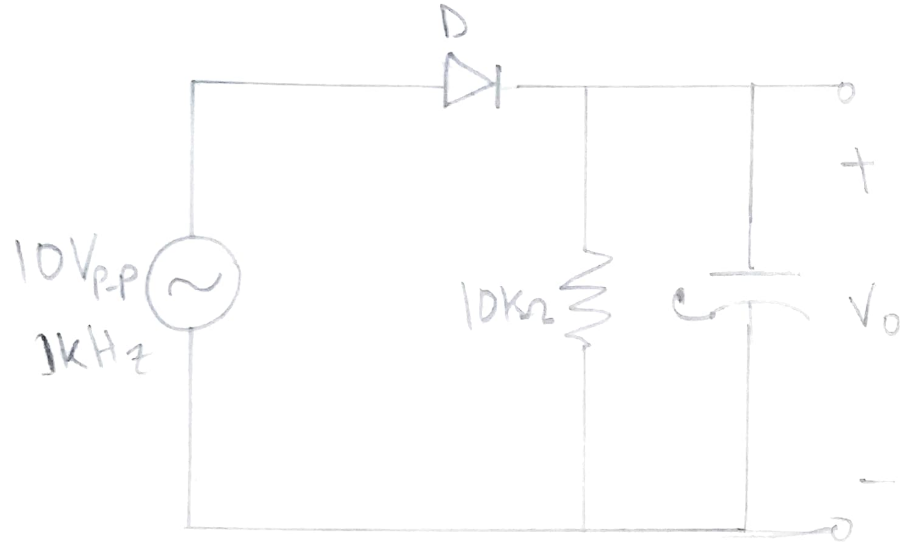


Figure: Experimental Circuit -1

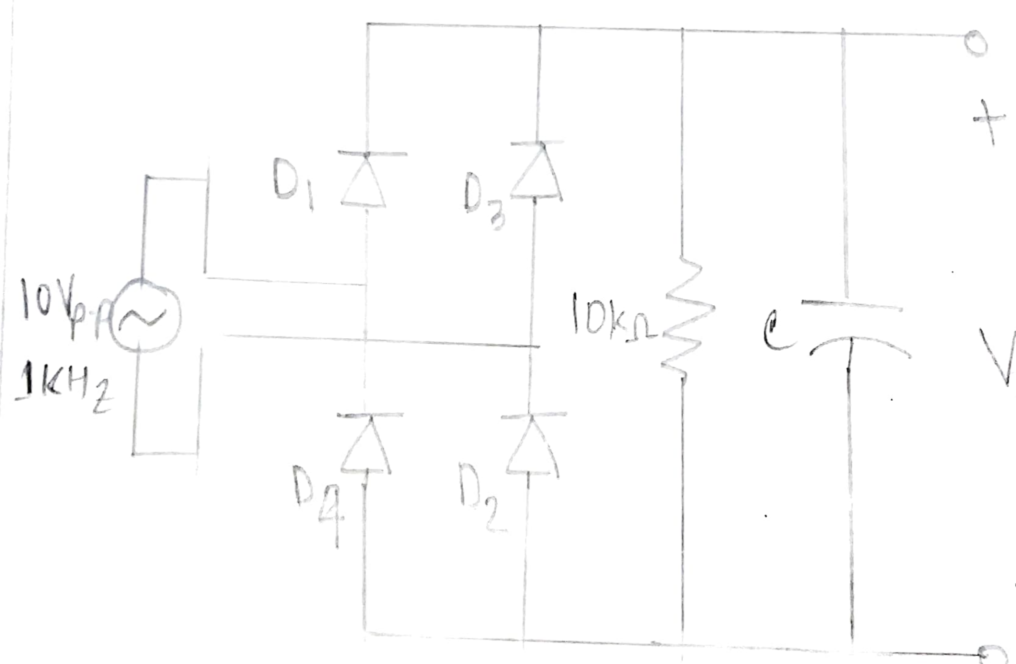


Figure: Experimental Circuit- 2

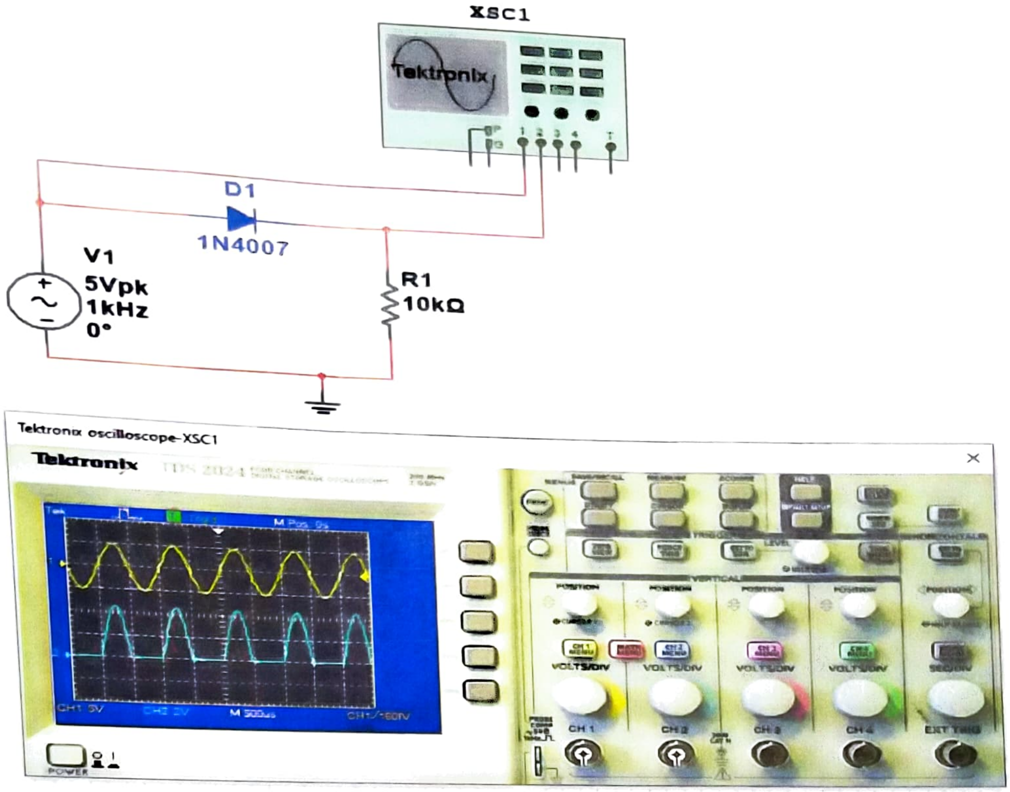


Fig –1: Half–wave Rectifier circuit with no capacitor

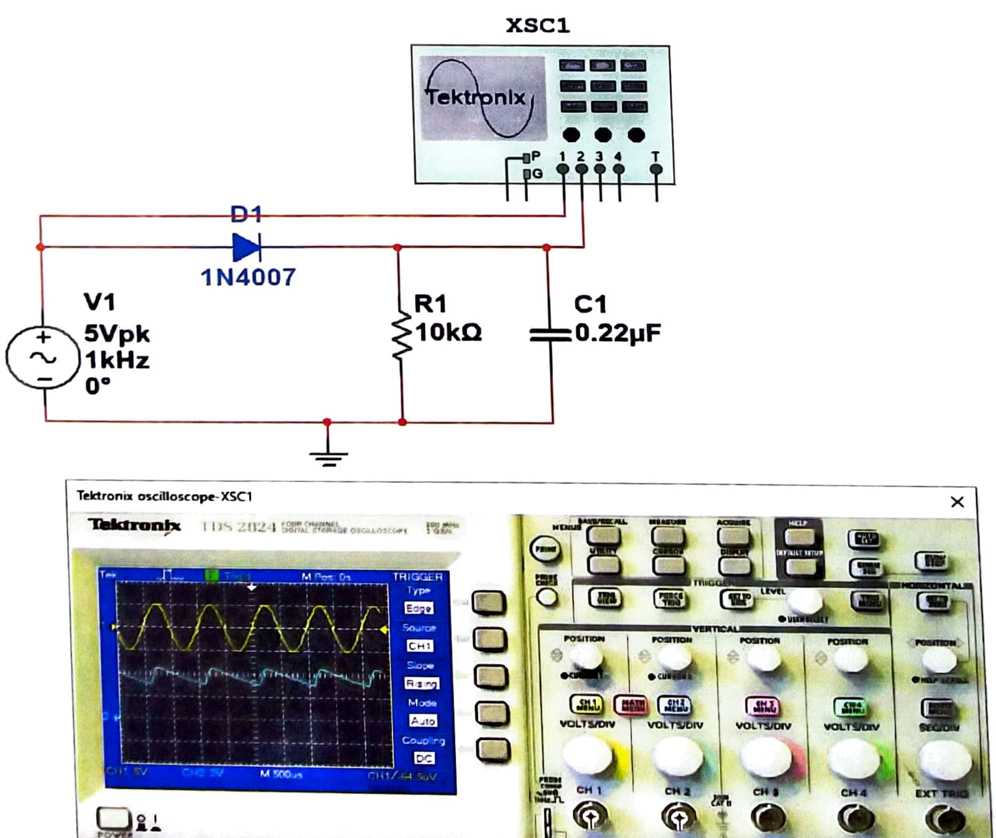


Fig –2: Half–wave Rectifier circuit with 0.22µF capacitor

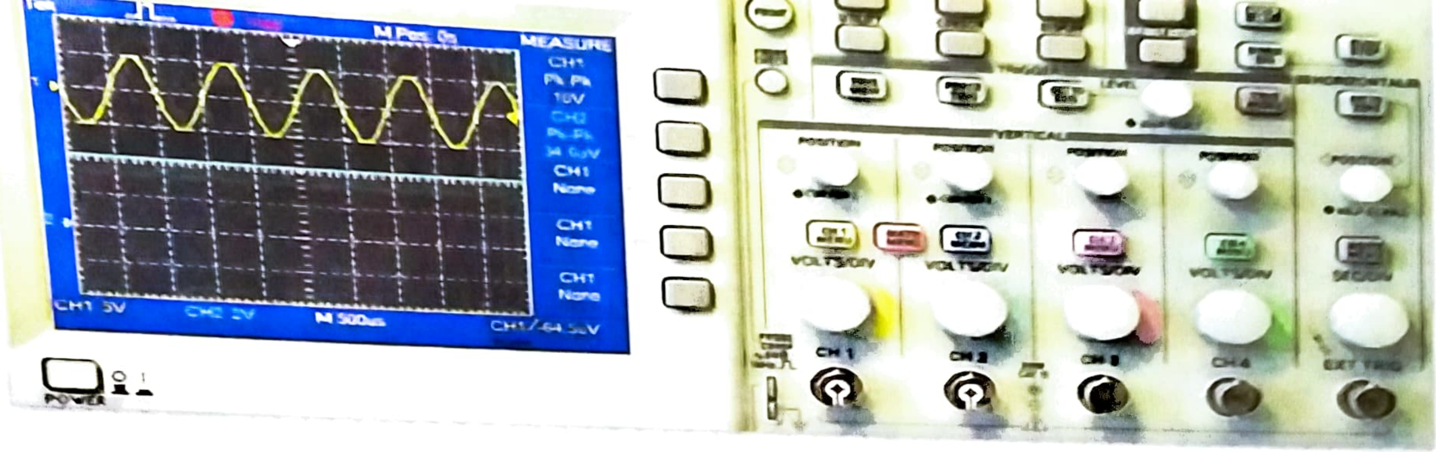
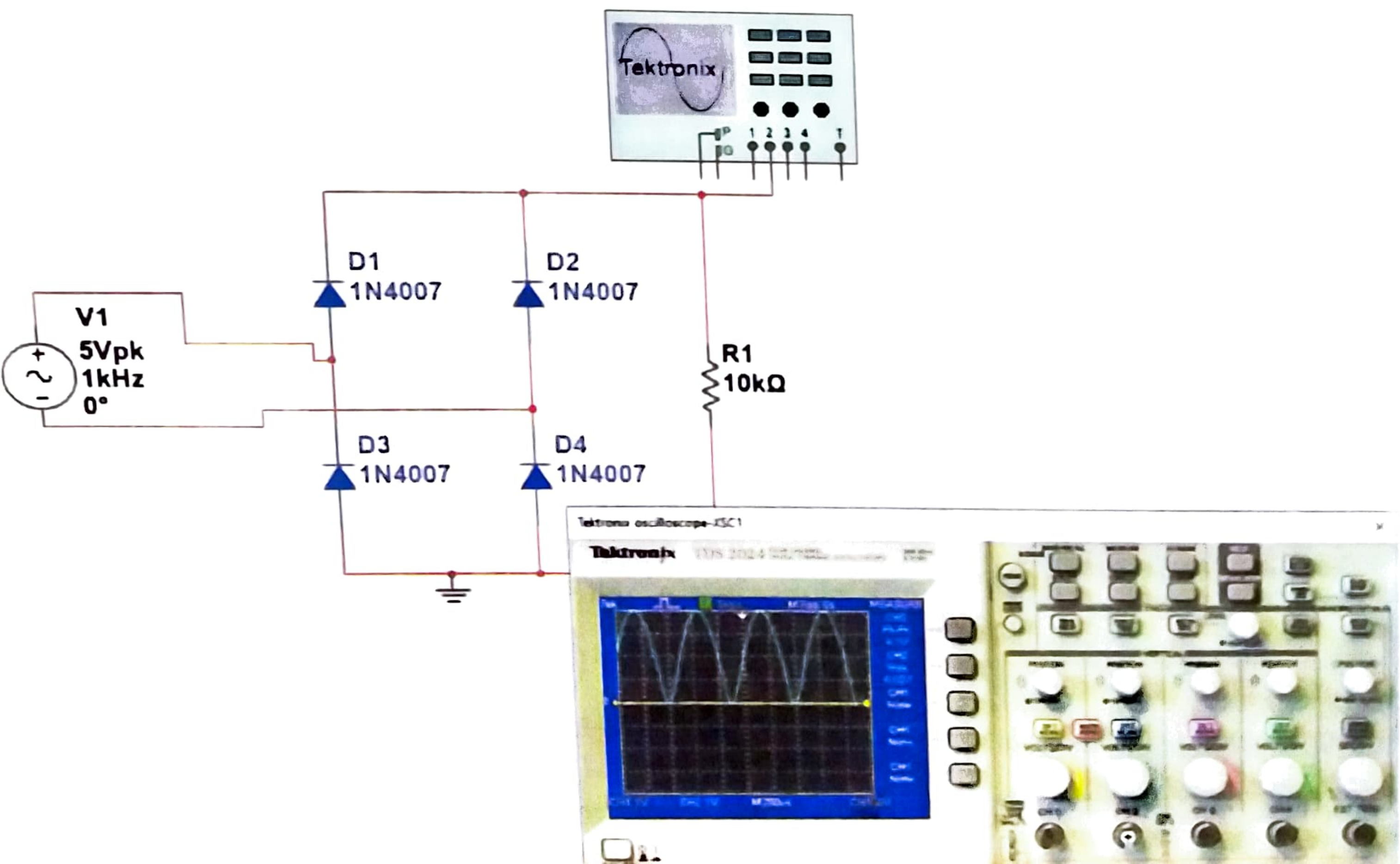


Fig –3: Half–wave Rectifier circuit with 10μF capacitor



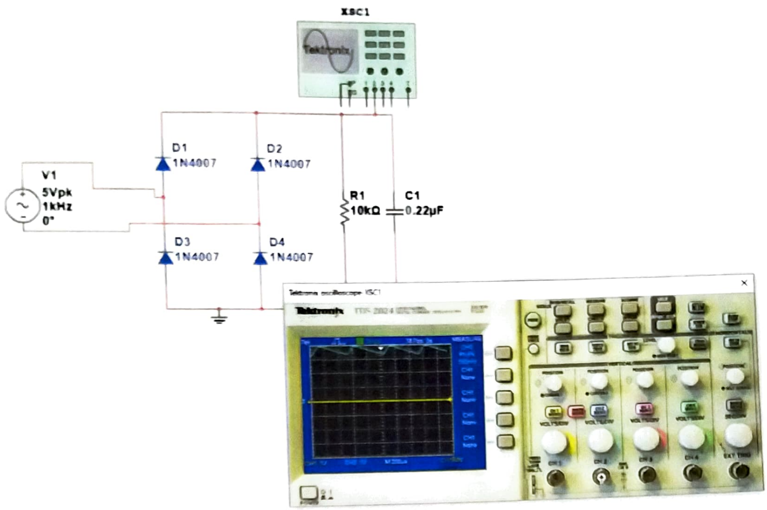


Fig –5: Full-wave Rectifier circuit with 0.22µF capacitor

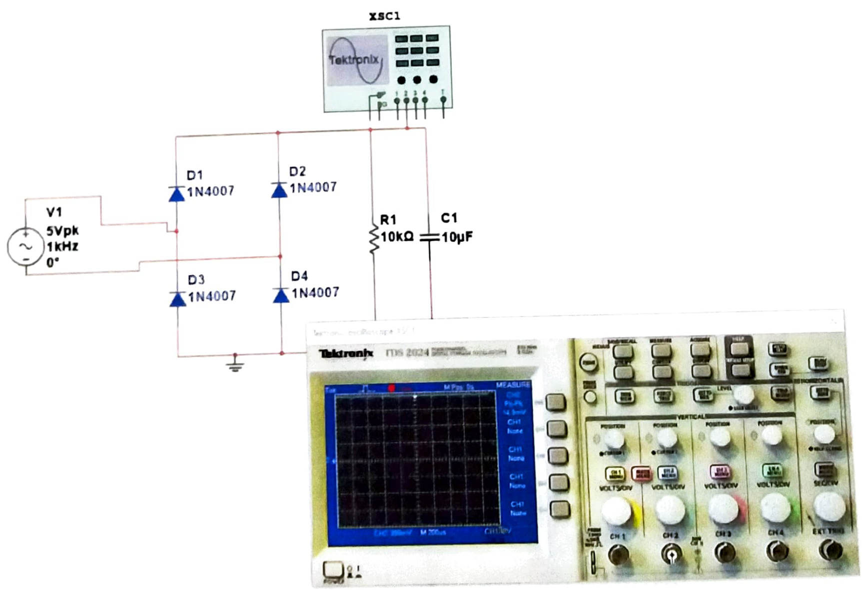


Fig –6: Full-wave Rectifier circuit with 10µF capacitor

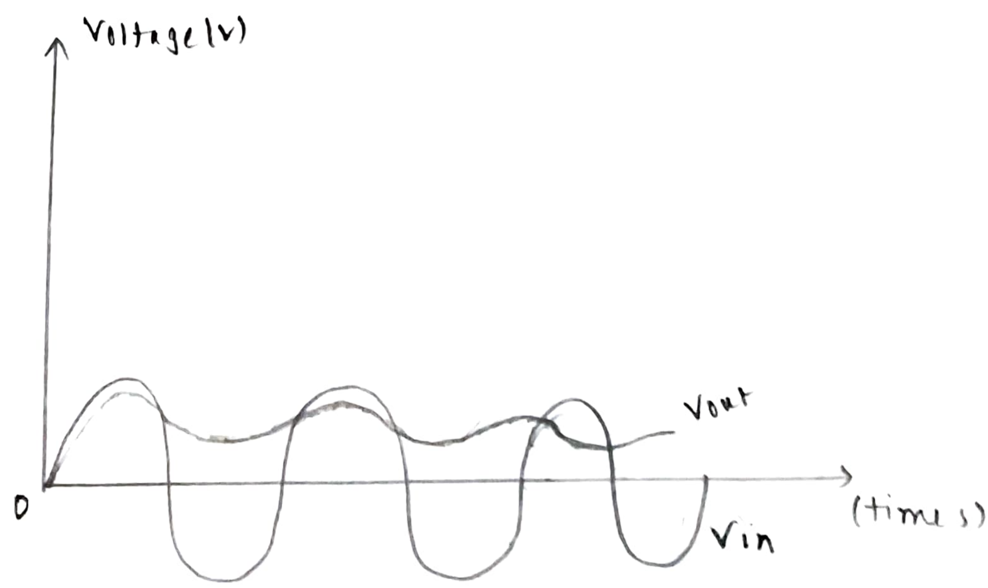
Question Answer:

(1) In the half wave, rectifier when I changed the capacitor to 0.22 μF to 10 μF, the output graph changed a lot. The discharging time for the μF capacitor was bigger hence the graph for the 10 μF capacitor was much straighter and the 0.22 μF capacitor graph was sort of a wave.

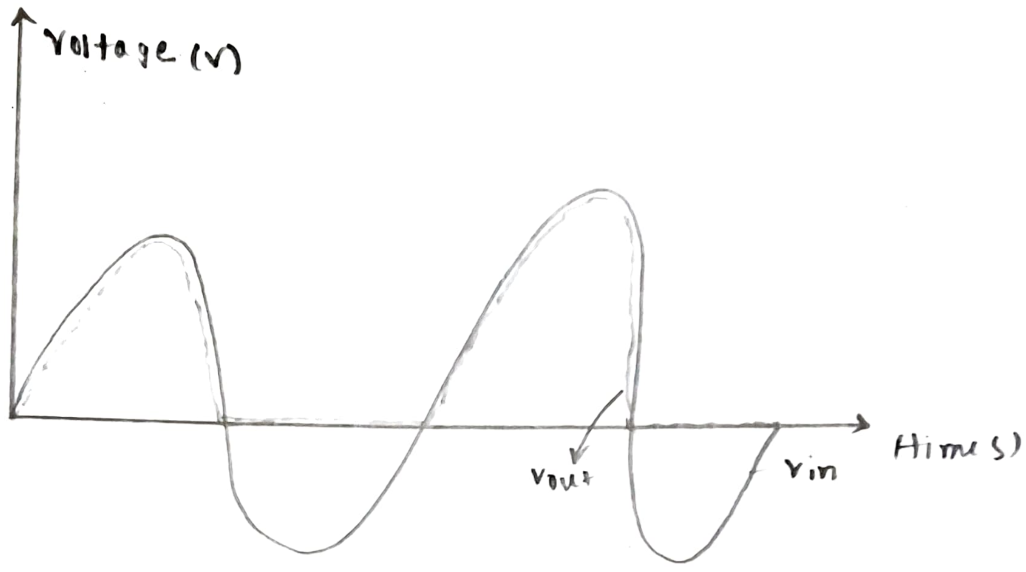
When the frequency is changed from 100 Hz to 1 KHz to 10 KHz, I can see that the graph is getting straighter as the frequency increases and the peak voltage decreases also.

The same thing happens for the full wave rectifier as well. The graph for 10 μF is much straighter than 0.22 μF and as the frequency increases and the peak voltage decreases also.

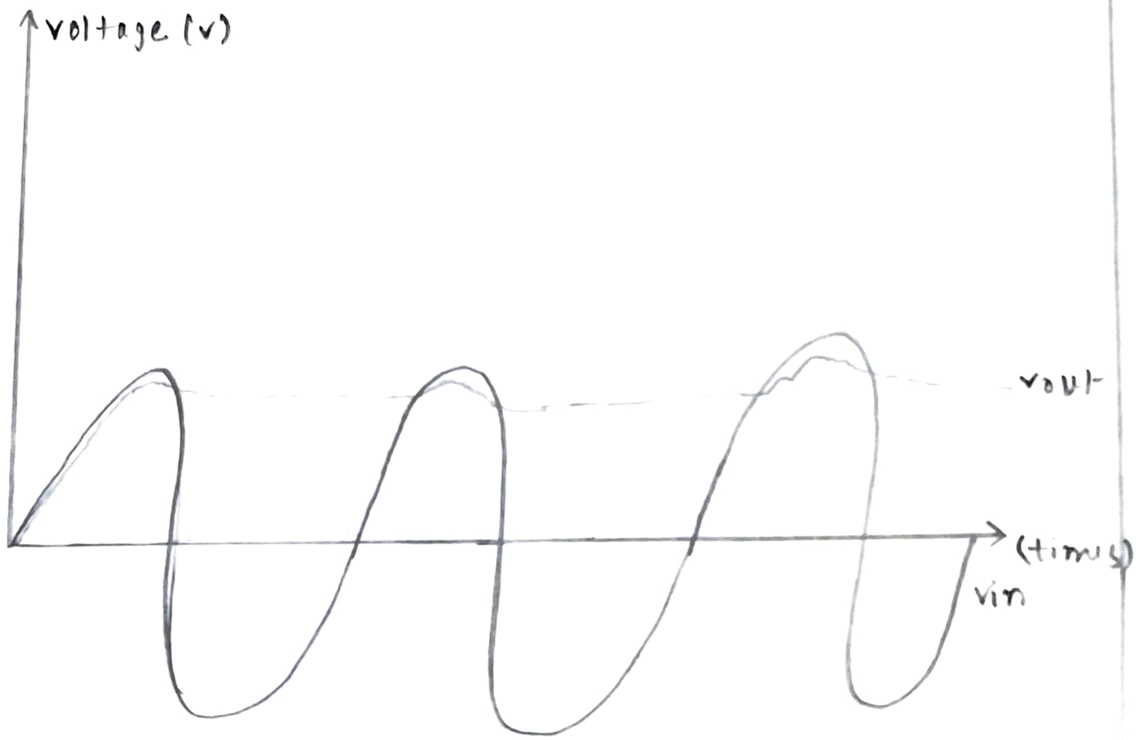
(2) For half wave rectifier circuit: (with 10pF capacitor)



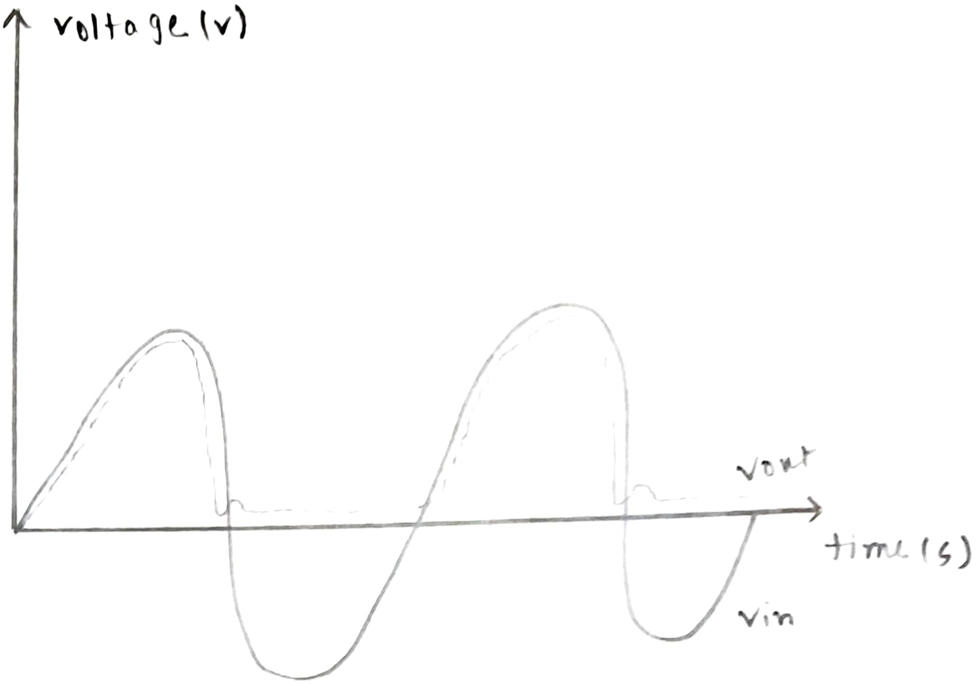
without capacitor,



Now, for full wave rectifier circuit (with 10pF capacitor)



without capacitor,



(3) If we change the input signal frequency, for half wave rectifier circuit, we will get equal supply as input frequency and double for full wave rectifier. Frequency is measured by how frequency the period is completed in one second. A time period (denoted by 'T') is the time needed for one complete cycle of vibration to pass in a given point. The output signal completes a period equal or twice as fast as the input frequency, as we can see in the graph. But, for 10 µF capacitor, we will get 0 frequency as the wave does not depend on time anymore.

(4) The functionality of the capacitor is to store charge and when the frequency is decreasing it will release the stored charge in it. The higher the capacitor the higher will be the capacitance of the capacitor. Hence 10 μF stores more than 0.22 μF. Due to storing more charge the discharge time of 10 μF is greater than 0.22 μF.

Discussion :

In this experiment, we had to study and analyze the Diode rectifier circuits. ~~IV characteristics of diode~~. We leart about the working of ractifiers. There are 2 types of rectifiers : Half wave and Full wave. They have their own characteristics. In a full wave rectifier the Positive half cycle follows a Path and is turned into DC current and the negetive half cycle goes to another Path and is Converted into DC current. The experiment was quite simple and the simulation was easy to conduct. The data and graph was exactly how we expected. Luckily we didn't face any kind of Problem. Everything gone well.

Experiment No: 02

Name of the Experiment: Diode rectifier circuits.

Objective:

Study of different diode rectifier circuits.

Theory:

A rectifier converts an AC signal into a DC signal. From the characteristic curve of a diode we observe that if allows the current to flow when it is in the forward bias only. In the reverse bias it remains open. So, when an alternating voltage (signal) is applied across a diode it allows only the half cycle (positive half cycle depending on the orientation of diode in the circuit) during its forward bias condition, other half cycle will be clipped off. In the output the load will get DC signal.

Diode rectifier can be categorized in two major types. They are -

1. Half-wave rectifier.
2. Full-wave rectifier.

Half - Wave Rectifier: Half-wave rectifier can be built by using a single diode. The circuit diagram and the wave shapes of the input and output voltage of half wave rectifier are shown bellow (figure 2.1) -

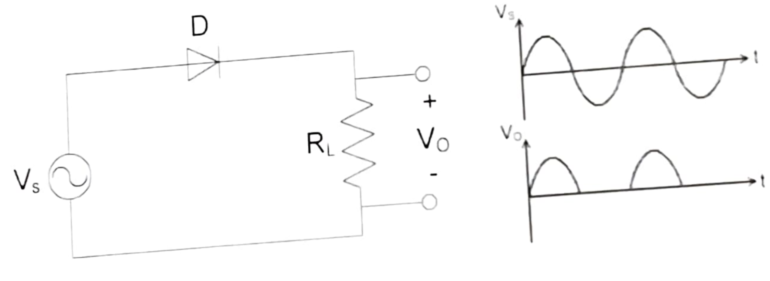


Figure 2.1: Half Wave Rectifier

The major disadvantages of half wave rectifier are -

- In this circuit the load receives approximately half of input power.
- Average DC voltage is low.
- Due to the presence of ripple output voltage is not smooth one.

Full Wave Rectifier: in the full-wave rectifier both the half cycle is present in the output. Two circuits are used as full-wave rectifier are shown bellow -

a) Full-wave rectifier using center-tapped transformer.
b) Full-wave bridge rectifier.

Full-wave rectifier using center-tapped transformer: two diodes will be connected to the ends of the transformer and the load will be between the diode and center tap. The circuit diagram and the wave shapes are shown in bellow (figure 2.2) -

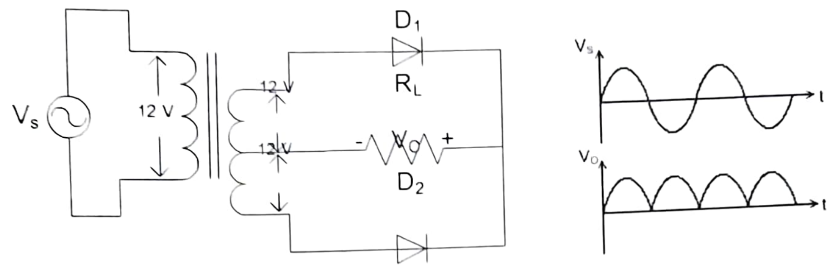


Figure 2.2: Full Wave Rectifier Using Center Tapped Transformer.

Full-wave rectifier using center-tapped transformer circuit has some advantages over full-wave rectifier. Those are -

- Wastage of power is less.
- Average DC output increase significantly.
- Wave shape becomes smoother.

The disadvantages of full-wave rectifier using center-tapped transformer are -

- Require more space and becomes bulky because of the transformer.
- Not cost effective (for using transformer).

Full-wave bridge rectifier: a bridge rectifier overcomes all the disadvantages of described above. Here four diodes will be connected as bridge connection. The circuit diagram and the wave shapes are shown in bellow (figure 2.3) -

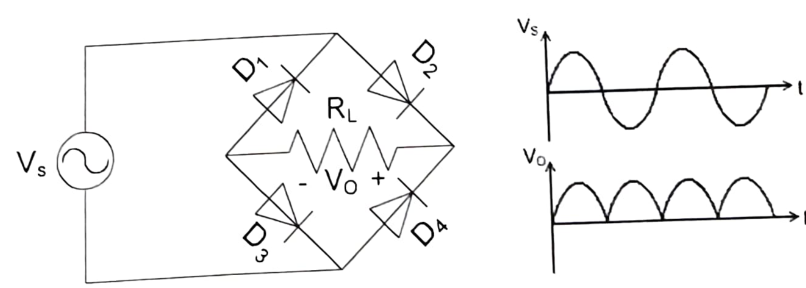


Figure 2.3: Full Wave Bridge Rectifier.

This rectifier however cannot produce a smooth DC voltage. It produces some ripple in the output. This ripple can be reducing by using filter capacitor across the load.

Serial no.	Component Details	Specification	Quantity
1.	p-n junction diode	1N4007	4 piece
2.	Resistor	10KΩ	1 piece
3.	Capacitor	0.22µF, 10µF	1 piece each
4.	Signal generator		1 unit
5.	Trainer Board		1 unit
6.	Oscilloscope		1 unit
7.	Digital Multimeter		1 unit
8.	Chords and wire		as required

Experimental Setup:

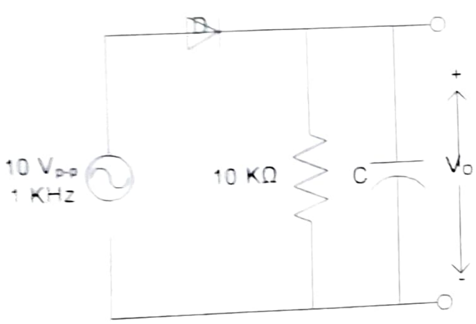


Figure 2.4 : Experimental Circuit 1.

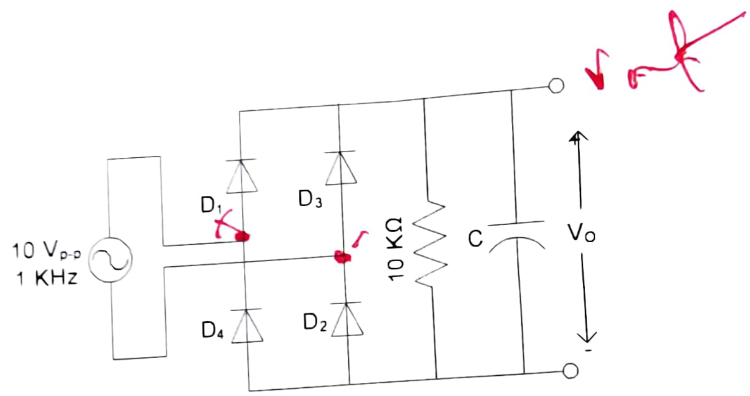


Figure 2.5 : Experimental Circuit 2.

Procedure:

1. Connect the circuit in breadboard as shown in figure 2.4 without capacitor.
2. Observe the output and input voltages in the oscilloscope and draw them.
3. Connect the 0.22µF capacitor and repeat step 2.
4. Connect the 10µF capacitor and repeat step 2. How does the output wave-shape differ from that in step 3?
5. Vary the frequency from 10 KHz to 100 Hz. What effects do you observe when frequency is changed?
6. Connect the circuit breadboard as shown in figure 2.5 without capacitor.
7. Observe the output and input voltages in the oscilloscope and draw them.
8. Connect the 0.22µF capacitor and repeat step 7.
9. Connect the 10µF capacitor and repeat step 7. How does the output wave-shape differ from that in step 8?
10. Vary the frequency from 10 KHz to 100 Hz. What effects do you observe when frequency is changed?

Report:

1. Write the answers that were asked during the working procedure.
2. Draw the input wave, output wave (without and with capacitor) for both the circuits.
3. What is the effect in output for changing input signal frequency for both the circuits (without and with capacitor)?
4. What is the function of capacitor in the both circuits? Why a capacitor of higher value is preferable?

Input and Output Waveform of the Half Wave Recti